

Vishal Kumar Rai

Ph.D. Candidate, Economics
Structural Macroeconomics & Life-Cycle Modeling
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WORKING PAPERS

Climate, Pollution and Endogenous Retirement

Develops a continuous-time life-cycle model with endogenous retirement to study how environmental degradation affects mortality, productivity, and retirement decisions.

Endogenous Retirement in the Presence of a Pandemic

Examines retirement, consumption, and health investment decisions under pandemic-induced mortality risk using a life-cycle model with epidemiological shocks.

Occupational Variation in Work Strain: A Life-Cycle Model of Retirement

Examines how occupation-specific work strain affects health investment and retirement decisions in a heterogeneous life-cycle framework.

RESEARCH INTERESTS

Structural Macroeconomics • Life-Cycle Economics • Aging and Retirement

EDUCATION

Ph.D. in Economics, Indian Statistical Institute – Delhi (Expected 2026)

Thesis: Essays on Life-Cycle Decisions under Environmental and Epidemiological Risks

M.A. in Economics, Delhi School of Economics 2019

B.A. (Hons.) Business Economics, University of Delhi 2015

SKILLS

Programming: Julia, Python **Methods:** Dynamic Programming, Structural Modeling

Data: MEPS, CPS, CDC Life Tables **Tools:** L^AT_EX, Git

CONFERENCE PRESENTATIONS

Endogenous Retirement in the Presence of a Pandemic

CDE–IEDS International Conference on Economic Development, Demographic Transition and Environmental Sustainability, Delhi School of Economics, June 2026.

EXPERIENCE & TEACHING

Teaching Assistant, ISI Delhi (Prof. Chetan Ghate)

Macroeconomics I & Global Macroeconomics (MSQE and Ph.D. coursework)

Tutorials, grading, and course coordination.

COMPUTATIONAL TOOLS

SwitchingControl.jl — Registered Julia package for solving endogenous regime-switching optimal control and dynamic economic models using event-driven differential equation methods.

REFERENCES

Monisankar Bishnu (Supervisor)

Professor,
Economics and Planning Unit
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Chetan Ghate

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